

Fit-Scorer Validation Study

Deliverable #2 / Success Metric #1 · 20 June 2026 · Targets: top-1 ≥ 70%, top-3 ≥ 90%

What was measured

The falsifiable proposal claim: *a resume scores highest against postings from its own field*. Each resume is scored against one synthetic requirement profile per job family (built from the LinkedIn postings corpus), then families are ranked. The harness scores with the **live engine** (it imports the real parseResume / parseJD / scoreFit / embeddings), so the number reflects the shipping product.

Corpus: 2,484 resumes → 2,362 after dropping the 3 under-populated families (BPO, AUTOMOBILE, AGRICULTURE) and corrupt rows → **21 families**. Field profiles from 48,069 postings mapped via a curated title→family map (43% title coverage).

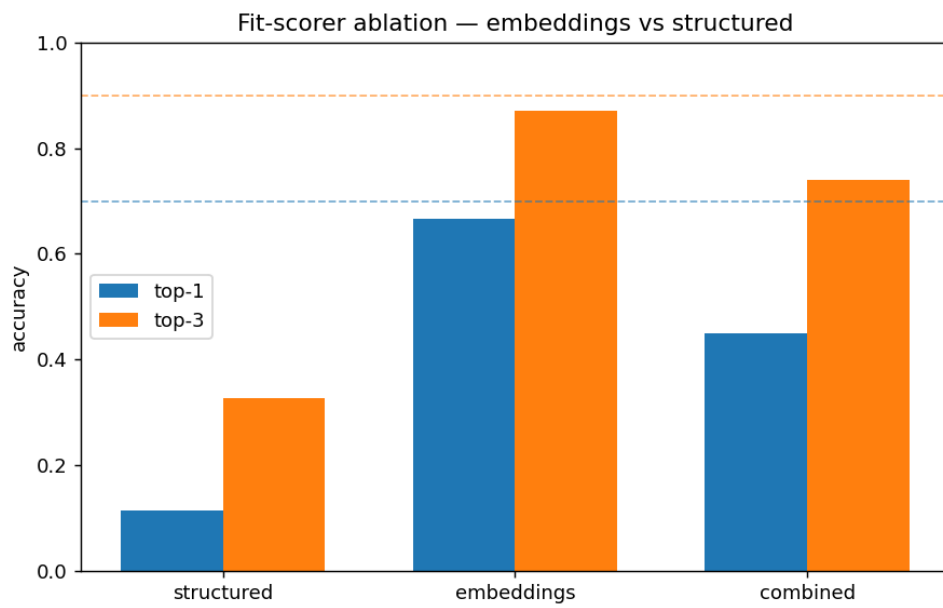
Result — and the headline finding

Arm	top-1	top-3	Notes
structured — full corpus (live engine)	12.1%	32.6%	rules + O*NET only
structured — subset	11.4%	32.7%	same 840 resumes as below
embeddings — subset (real BGE)	66.5%	87.0%	nearly hits target out of the box
combined — subset (50/50)	45.0%	73.9%	naive blend; weak arm drags it down
targets	70%	90%	

The missing half of the proposal's own methodology — "semantic embeddings *plus* deterministic rules" — is the half that works. Real BGE-small embeddings take top-1 from ~12% to **66.5%** and top-3 to **87.0%**, within striking distance of the 70/90 targets, while the shipped rules-only engine flatlines near chance (21-class random ≈ 4.8%). The naive 50/50 blend hurts, so the blend must weight embeddings higher. Embeddings were never wired into the live fit engine — that is the top lever to clear target.

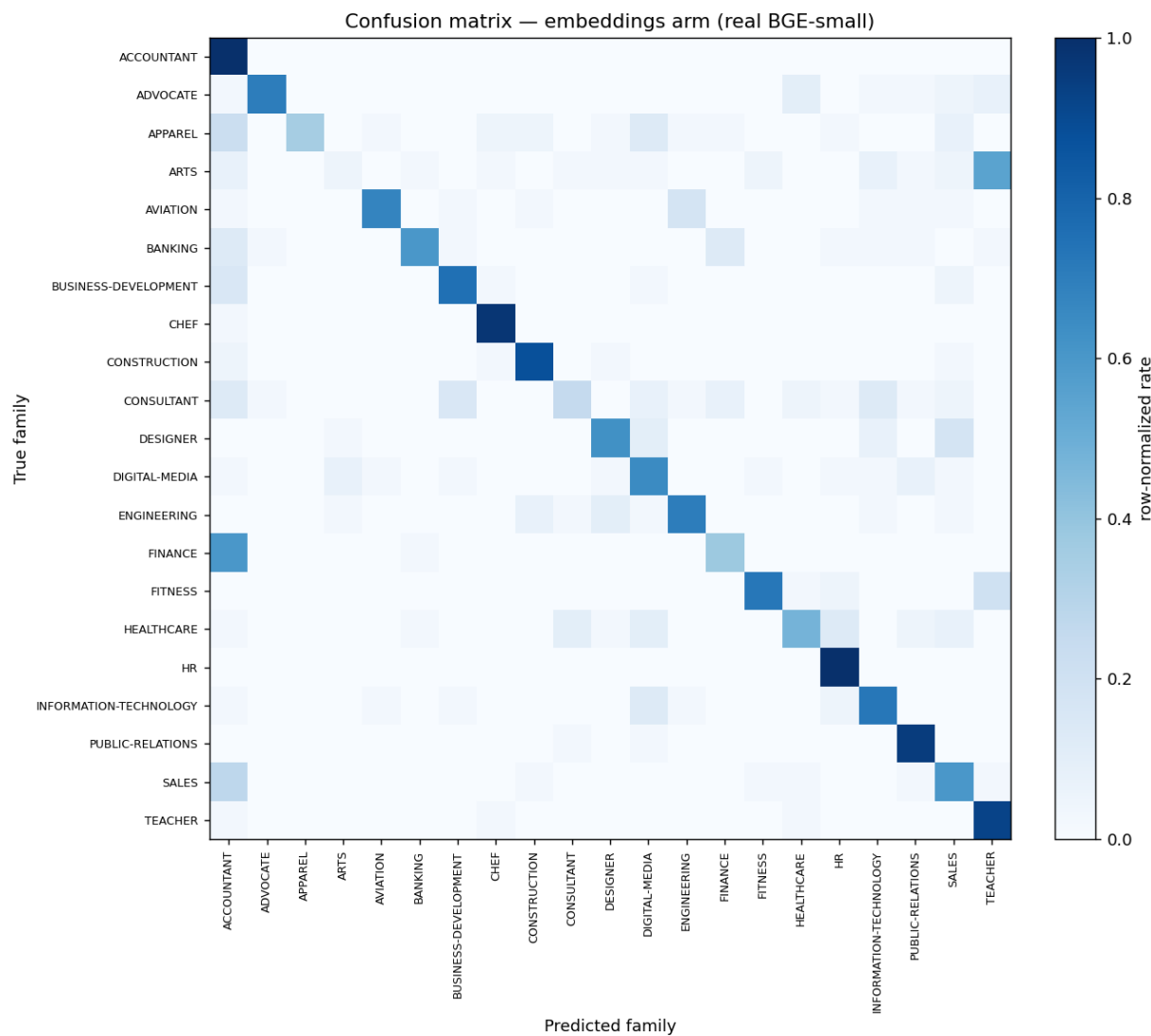
Embeddings/combined figures are a stratified 40/family subset (840 resumes); the structured full-corpus number (2,362) is shown for reference. The harness scores 100% on clean fixtures, confirming the low structured number is an engine/methodology limit, not a harness bug.

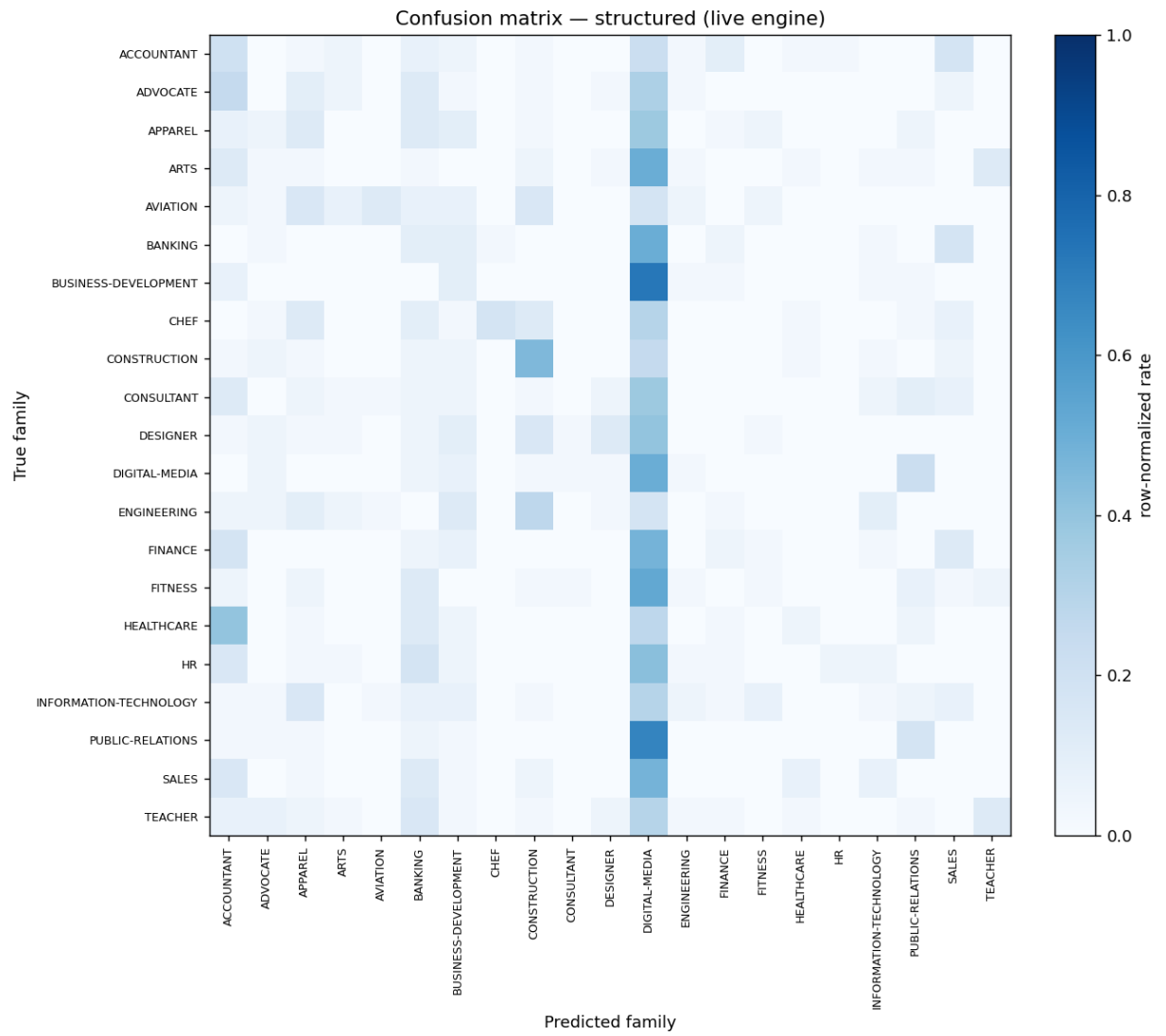
Ablation — embeddings vs structured



Confusion matrices

Row-normalized (true family → predicted family). The embeddings arm has a strong diagonal; the structured arm scatters off-diagonal.





Per-family top-1 (structured arm)

Family	n	top-1
DIGITAL-MEDIA	40	50.0%
CONSTRUCTION	40	45.0%
ACCOUNTANT	40	20.0%
CHEF	40	17.5%
PUBLIC-RELATIONS	40	17.5%
APPAREL	40	12.5%
AVIATION	40	12.5%
DESIGNER	40	12.5%
TEACHER	40	12.5%
BANKING	40	10.0%
BUSINESS-DEVELOPMENT	40	10.0%
FINANCE	40	5.0%
HEALTHCARE	40	5.0%
HR	40	5.0%
FITNESS	40	2.5%
INFORMATION-TECHNOLOGY	40	2.5%
ADVOCATE	40	0.0%
ARTS	40	0.0%
CONSULTANT	40	0.0%
ENGINEERING	40	0.0%
SALES	40	0.0%

Lever to reach target

- **Add the embeddings arm to the live fit engine** (`lib/matching.ts`) and weight the blend toward embeddings — it nearly clears target alone.
- Expand the O*NET taxonomy beyond 21 occupations (the structured arm is tech/analytics-biased).
- Improve the postings title→family mapping coverage (currently 43%).

Reproduce

```
npm run validate:prep → npm run validate:fit → npm run validate:report. Real embeddings: npm i @xenova/transformers then EMBEDDINGS_ENABLED=true npm run validate:fit -- --sample 40. Smoke (no dataset): npm run validate:smoke. Harness: scripts/validation/.
```